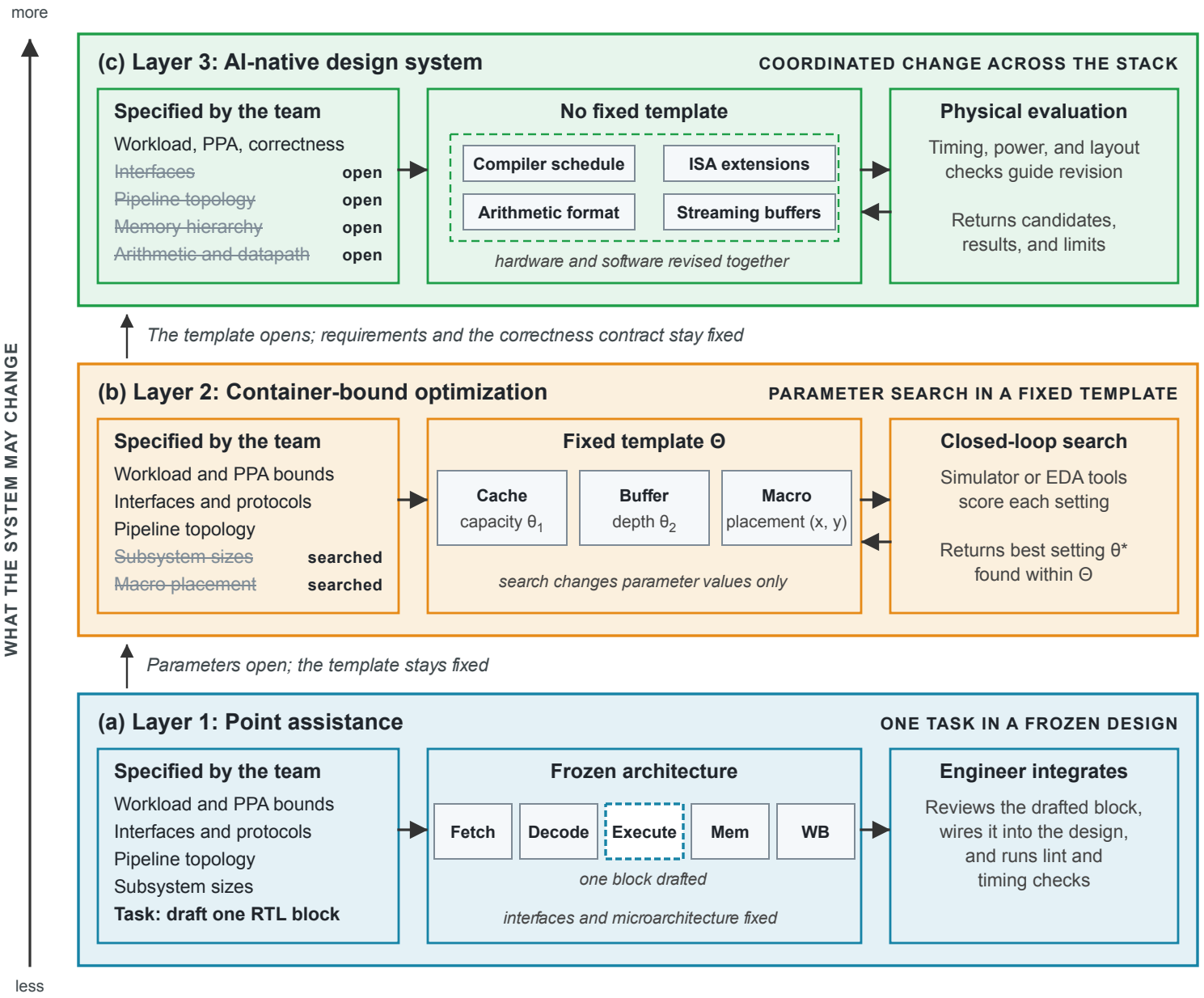

more
WHAT THE SYSTEM MAY CHANGE
less
(c) Layer 3: AI-native design system
COORDINATED CHANGE ACROSS THE STACK
Specified by the team
Workload, PPA, correctness
Interfaces open
Pipeline topology open
Memory hierarchy open
Arithmetic and datapath open
No fixed template
Compiler schedule
ISA extensions
Arithmetic format
Streaming buffers
hardware and software revised together
Physical evaluation
Timing, power, and layout checks guide revision
Returns candidates, results, and limits
The template opens; requirements and the correctness contract stay fixed
(b) Layer 2: Container-bound optimization
PARAMETER SEARCH IN A FIXED TEMPLATE
Specified by the team
Workload and PPA bounds
Interfaces and protocols
Pipeline topology
Subsystem sizes searched
Macro placement searched
Fixed template Θ
Cache capacity θ1
Buffer depth θ2
Macro placement (x, y)
search changes parameter values only
Closed-loop search
Simulator or EDA tools score each setting
Returns best setting θ* found within Θ
Parameters open; the template stays fixed
(a) Layer 1: Point assistance
ONE TASK IN A FROZEN DESIGN
Specified by the team
Workload and PPA bounds
Interfaces and protocols
Pipeline topology
Subsystem sizes
Task: draft one RTL block
Frozen architecture
Fetch
Decode
Execute
Mem
WB
one block drafted
interfaces and microarchitecture fixed
Engineer integrates
Reviews the drafted block, wires it into the design, and runs lint and timing checks